# PHAS0050 : Climate & Energy
# Module 1 : Climate Science

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# 1 Introduction

The aim of this part of the course is to review the following topics:

- The climate record and evidence for anthropogenic climate change.
- The IPCC reports and radiative forcing estimates.
- The global carbon cycle.
- Influence of atmospheric gases, clouds and aerosols on climate.
- The global energy balance including solar variation, and photon absorption, scattering and re-radiation from components of the atmospher.
- Current methods for modelling the global climate: likely uncertainties. Macro- and microscopic scale modelling.

The lecture slides on Climate Science (available on Moodle) cover this material; these notes provide some additional detail in support of those slides. Although it goes into significantly more detail than required for this course, the best introductory level textbook for those wishing to delve in more detail into the fundamental physics is [1].

# 2 The Climate Record

We have direct instrumental records of atmospheric and ocean temperatures going back to around 1850, and atmospheric composition data for a shorter period than that. How, then, do we know that recent changes are not just a typical fluctuation and nothing to be concerned about? The answer lies in a number of geological and geophysical records, and in particular in ice cores taken from the Antarctic and Greenland ice caps. Gas bubbles in the ice provide a record of the chemical state of the atmosphere going back over 600 kyr. Deducing the ancient temperature is harder though, and relies on the slightly different evaporation/condensation behaviour of water that contains heavy isotopes of hydrogen or oxygen. The oceans and the ice caps effectively operate a massive distillation process, in which the transport of heavy water molecules occurs at a slightly lower rate, such that ice caps have a higher concentration of light water than the oceans. However, this differential behaviour is temperature dependent, such that at higher ocean and polar atmospheric temperatures the depletion of heavy isotopes is reduced. Thus the $^{18}\mathrm{O}$ and deuterium concentrations in polar ice are positively correlated with mean global temperature. Establishing a numerical correlation requires a comparison between recent ice core properties and directly measured temperatures, and on the basis of this, one can go on to deduce the ancient temperature record. In addition, there is a wealth of ancient biological data that can serve as a proxy for temperature, including the width of tree rings and shell growth patterns.

# 3 The Natural Greenhouse Effect

It is important to recognise that the greenhouse effect is not anthropogenic. It is natural and provides us with a considerable warming that makes life on Earth comfortable for us and for much of the biosphere. It is the *enhanced* greenhouse effect that is anthropogenic. It should be appreciated that part of the difficulty in understanding climate change is that humankind is making a small perturbation to an already complex system.

We start by considering the balance between incident radiation from the sun and black body radiation from the Earth, under the assumption that the atmosphere does *not* participate in the radiative transfer process. The incident radiative flux, seasonally and globally averaged, is $342\ \mathrm{W/m^2}$. Some of this is reflected

from the surface: the planetary albedo $A$ represents this fraction. Let us estimate this to be 15%: the true Earth albedo is more than 30%, but this includes reflection from clouds, and here we are ignoring this contribution. So the temperature $T_e$ of the terrestrial black body radiation is given by $\sigma T_e^4 = 342 \times 0.85\ \mathrm{W/m^2}$, where $\sigma$ is the Stefan-Boltzmann constant, and thus $T_e \approx 267$ K. The actual mean terrestrial surface temperature is some 20 degrees higher than this.

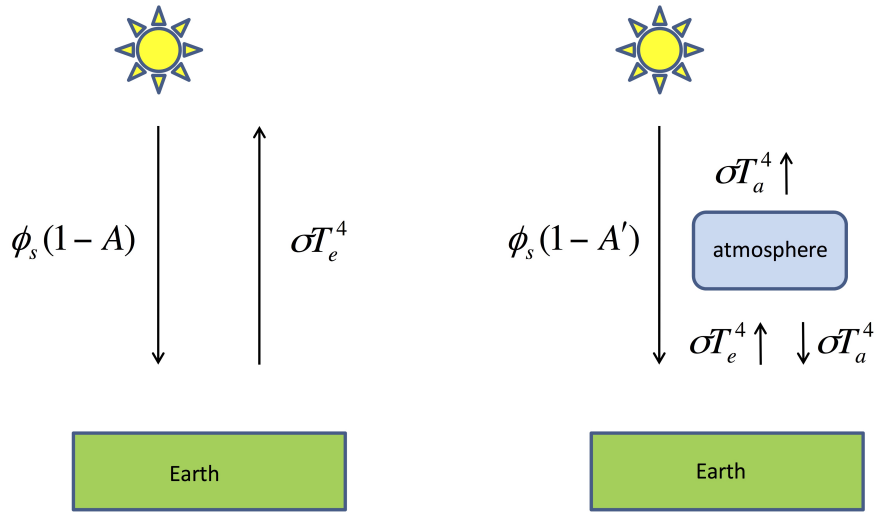


Figure 1: Simple model of the greenhouse effect. Radiative transfer for the Earth system with no participation by the atmosphere is contrasted with the more realistic, but still simplified, situation on the right where greenhouse gases, aerosols and clouds interact with the terrestrial radiation.

The following analysis provides a very simple demonstration of this warming. Consider the radiative balance of a system whereby all the terrestrial radiation is intercepted by an atmospheric blanket, and re-emitted as black body radiation at a temperature $T_a$. Half of this emission is directed back towards the Earth, and half out into space, as illustrated in Figure 1. The latter part must balance the incoming solar radiation, which should now be regarded as $342\ \mathrm{W/m^2}$ reduced by an enhanced planetary albedo, taking into account additional reflection off the atmosphere. We shall use 30%, such that $\sigma T_a^4 = 342 \times 0.7\ \mathrm{W/m^2}$, in which case $T_a \approx 254$ K. The surface temperature, on the other hand, is determined by balance between the gain and loss of energy, by radiation, of the atmosphere:

$$\sigma T_e^4 = 2\sigma T_a^4$$

in which case $T_e = 2^{1/4} T_a \approx 303$ K. In this example, a gain of over 30 K has been provided by atmospheric participation in the radiative balance. Such a simple model cannot be expected to be fully accurate, of course, since it has neglected many important details. A significant flux that is absent in the analysis is the 10% of terrestrial radiation that is not intercepted by the atmosphere and is emitted directly to space, according to present estimates. This is the component of radiative transfer that is interfered with by the enhanced presence of greenhouse gases due to humankind's activities.

## 4 Active radiative components in the atmosphere

It is well known that greenhouse gases are active in atmospheric radiative transfer, most importantly in modifying the outgoing, infra-red terrestrial radiation as discussed above. They do so by virtue of the existence of transitions between vibrational and rotational molecular modes, which correspond to energies that match photons in the IR spectrum. Such gases include water vapour, carbon dioxide, methane and others. Energy that is absorbed is re-emitted, but often over a different spectrum, to a rough approximation

a black-body spectrum characterised by the temperature of the emitting gas. Greenhouse gases are naturally removed from the atmosphere, largely by the oceans and biosphere, but reduction in the mean would occur over decadal timescales (though the lifetime of an individual molecule in the atmosphere might be shorther than this).

Clouds are also familiar objects in the atmosphere that interfere with radiation, this time across a broad range of frequencies, both in the IR and the visible. There is the additional mechanism of the reflection of radiation at the interface between the gas and condensed phase, and can be understood on the basis of classical electromagnetism and optics. Radiation experiences refraction within cloud droplets (or crystals), and hence deviation from the incident direction, as well as extinction by the processes of photon absorption and re-emission mentioned above.

Less well known are atmospheric particulates that are sub-micron in size and almost invisible to the human eye. They are perceptable sometimes as atmospheric haze. These are atmospheric aerosols. Typically each cubic cm of air contains 100-1000 such particles, ranging in size from a few nm up to 10 $\mu$m. Above this range size, a particle has an appreciable settling velocity and does not remain aloft for very long. Otherwise, the average atmospheric lifetime of an aerosol particle before removal by settling or by cleansing (scavenging) by rainfall is perhaps a week.

Aerosols are generated naturally and anthropogenically, either by direct injection (for example as a product of combustion) or by *in situ* condensation from gas-phase precursors such as sulphur dioxide, which forms sulphate. They absorb and scatter radiation across a broad spectrum, through a variety of mechanisms since they are remarkably diverse chemically. Most aerosol is composed of sulphate, nitrate and chloride salts, often in aqueous solution; insoluble materials such as black carbon (soot); or organic substances such as the oxidation products of the complex volatiles given off by vegetation. In addition there might be mineral dust elevated by the wind, or originating from meteor residues. The aerosol optical depth of the atmosphere, which is a measure of total extinction of radiation due to particulates, and is defined as the coefficient $\alpha$ in the relation $\phi_{tr} = \phi_{in} \exp(-\alpha)$ between incident flux $\phi_{in}$ and transmitted flux $\phi_{tr}$ , can be of order unity in certain regions of the world, at certain times. Such interference in solar radiation due to particulate haze has been called *global dimming* or the *direct* aerosol effect on climate. But in the next section we shall consider the so-called *indirect* aerosol effect, namely the way particulates control the properties of cloud.

## 5 Aerosols and cloud formation

A simplistic view of cloud formation is that water condenses when the temperature of the air falls below the local dew point. This is, as the name suggests, the temperature below which liquid condenses from the available water vapour: at the dew point the vapour and liquid phases are in equilibrium. If a certain number $N$ of water molecules are held in an air parcel, then the dew point is defined to be the temperature at which the vapour pressure $p_v = NkT/V$ is equal to the saturated vapour pressure, given approximately by the Clausius-Clapeyron equation $p_{vs} \propto \exp(-L/kT)$, where $L$ is the latent heat of evaporation per molecule. The relative humidity $RH$ is defined to be the ratio of $p_v$ to $p_{vs}$ written as a percentage. Condensation is favoured when $RH$ exceeds 100%.

However, cloud formation is not so easy as it sounds. In fact each cloud droplet needs to nucleate on a seed: a pre-existing aerosol particle. And not all aerosol particles will provide such a nucleation centre: it depends on their size and chemical content. In order to understand this we would need to develop a model of the effectiveness of aerosol as CCN: cloud condensation nuclei, but this is beyond the scope of this module. But clearly the character of cloud is determined by the number of aerosol particles in the precursor air parcel, and by their chemical composition. This matters because clouds consisting of small droplets reflect more radiation (because there is more surface area for a given mass of water) and last longer before raining out than clouds of larger droplets. This is the Twomey effect, also known as the cloud albedo effect or indirect aerosol effect on climate. And the density and character of atmospheric aerosols is affected by

human activities.

## 6 Radiative forcing

The importance of various drivers of climate change can be assessed by calculating a radiative forcing (RF) for each. This is defined to be the effective perturbation to the downward energy flux at the top of the troposphere caused by a specific change in atmospheric or surface condition. For example, the radiative forcing of a certain increase in carbon dioxide in the atmosphere is to reduce the amount of upward flux from terrestrial emissions, which is viewed as equivalent to an increase in downward flux. A positive RF is a warming influence, and greenhouse gases lie in this category. Additional aerosols and cloud layers in the atmosphere are harder to assess since their composition, size range and altitude affect their radiative properties. Black carbon aerosols are warming whilst white sulphate aerosols are cooling, at a certain altitude and particle size, due to their different absorptive material properties. Layers at high altitude, such as cirrus cloud, can be warming, since they obscure outgoing radiation from rather warmer components beneath them and re-emit at a lower temperature and flux. But low-lying cloud layers can be cooling, since their temperature is closer to the obscured radiative surface beneath, and their dominant effect is the reflection of incoming solar radiation. Studies have been conducted to quantify the radiative forcings arising from anthropogenic changes to the atmosphere, and although uncertainties are still quite large, it appears that the overall averaged influence of aerosol on climate, including their effect on cloud, is a negative forcing.

Most anthropogenic forcings are less than 1% of the incident solar radiation flux, and the overall radiative forcing combining all contributions is estimated to be within the range $1.13 - 3.33\ \mathrm{W/m^2}$ (this is a 90% uncertainty interval), with a central value of $+2.3\ \mathrm{W/m^2}$ [2].

## 7 Climate modelling

Climate science is immensely complicated and yet we need models that attempt to predict future behaviour: how has this been done? In short, roughly, but with a great effort to minimise uncertainties. Typical global climate models represent the atmosphere and oceans on 100 km spatial scales and many processes that take place on smaller scales have to be parametrised. Cloud content is an example: each atmospheric cell is characterised by a temperature and a water content, and the partition of that water between condensed and vapour phases is determined by consideration of the thermodynamics. Nevertheless, we know that real temperatures and humidity vary widely over such spatial scales, and that cloud coverage is not uniform. The solution is to fit models to data to define and employ an effective cloud coverage and an empirically derived mean cloud drop size. Choices of model parameters, including the pollutant emission rates which could be estimates, lead to uncertainties in the outcomes of calculations, which are typically forward projections of globally averaged temperature change. The situation at present is that a given scenario of anthropogenic activity may be associated with a range of possible climatic effects: for example an expected temperature rise of 2.1 K with a 'likely' range of uncertainty of 1.4-3.1 K if the atmospheric carbon dioxide level is stabilised at 450 ppm. Furthermore, local changes are expected to differ considerably from the mean.

## References

[1] "Elementary Climate Physics", F. W. Taylor, Oxford University Press (2005).

[2] Assessment Report 5 (AR5) of the Intergovernmental Panel on Climate Change (IPCC), Working Group I Summary for Policymakers :
`https://www.ipcc.ch/site/assets/uploads/2018/02/WG1AR5_SPM_FINAL.pdf`